Experiment Number: 1

Simulation and Comparative Analysis of FDMA, TDMA and CDMA: Resource Allocation, User Capacity and Mobile-Generation Context

Aim:To simulate and compare the basic resource-allocation methods used in FDMA,TDMA and CDMA, and examine how theirassumed user capacity and interference behaviour differ.

Program 1 — Capacity estimate for FDMA, TDMA and CDMA
Objective 1: To estimate and compare the illustrative number of users supported by FDMA, TDMA and CDMA using the specified system parameters.

CODE:

```
% Experiment 1: FDMA, TDMA and CDMA Comparison
% Program 1: Capacity Estimation

clear;
clc;
close all;

% System Parameters
totalBandwidthHz = 1e6;    % Total bandwidth = 1 MHz
fdmaChannelHz = 200e3;     % FDMA channel width = 200 kHz
tdmaSlots = 8;             % TDMA slots per frame
processingGain = 64;       % CDMA processing gain
targetSIRdB = 7;           % Target SIR in dB

% Calculate Capacity
fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);
tdmaUsers = tdmaSlots;

% Convert target SIR from dB to linear scale
targetSIR = 10^(targetSIRdB / 10);

% Estimate CDMA capacity
cdmaUsers = floor(1 + processingGain / targetSIR);

% Display Results
fprintf('FDMA, TDMA AND CDMA CAPACITY COMPARISON\n');
fprintf(' ---------------------------------- \n');
```

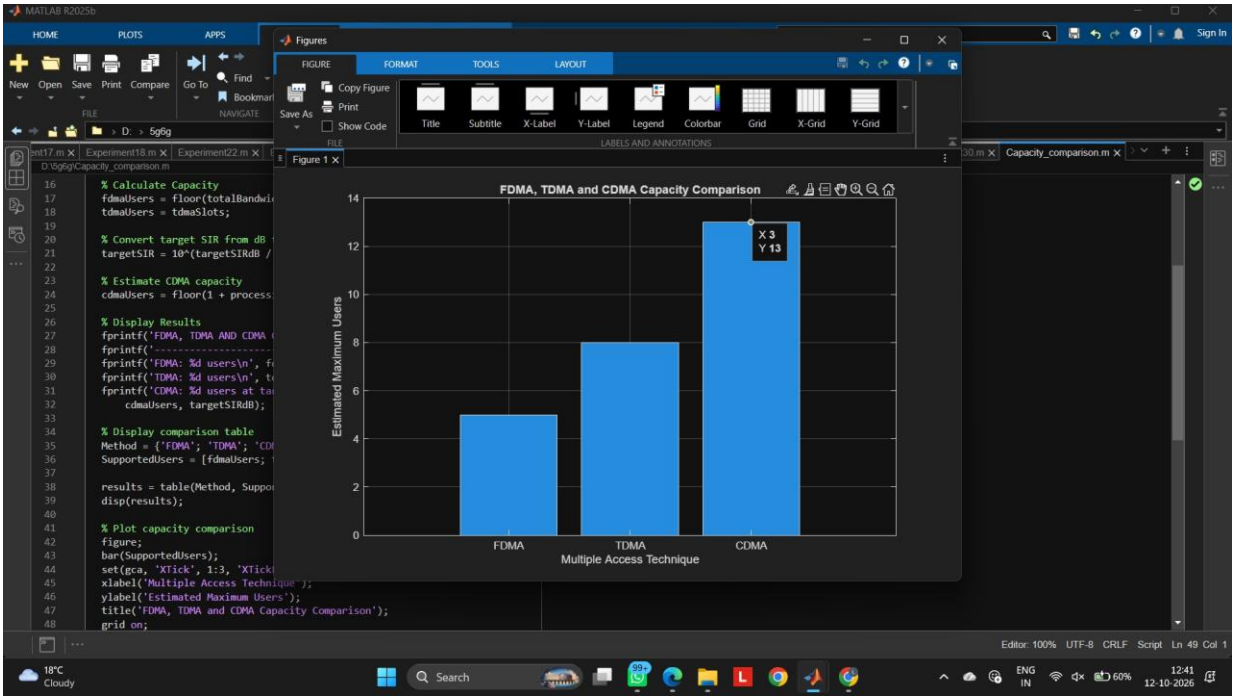
```
fprintf('FDMA: %d users\n', fdmaUsers);
fprintf('TDMA: %d users\n', tdmaUsers);
fprintf('CDMA: %d users at target SIR = %.1f dB\n', ...
    cdmaUsers, targetSIRdB);

% Display comparison table
Method = {'FDMA'; 'TDMA'; 'CDMA'};
SupportedUsers = [fdmaUsers; tdmaUsers; cdmaUsers];

results = table(Method, SupportedUsers);
disp(results);

% Plot capacity comparison
figure;
bar(SupportedUsers);
set(gca, 'XTick', 1:3, 'XTickLabel', Method);
xlabel('Multiple Access Technique');
ylabel('Estimated Maximum Users');
title('FDMA, TDMA and CDMA Capacity Comparison');
grid on;
```

OUTPUT:



Assessment Rubrics — Experiment 1: Simulation and Comparison of FDMA, TDMA and CDMA Assessment criterion Weight 4 — Strong 3 — Meets 2 — Partial Objective 1: Capacity estimation and comparison 10 Correct FDMA, TDMA and CDMA estimates; clear 1 — Limited Outputs mostly correct, with a minor error or incomplete explanation. Some correct outputs, but errors or omissions remain. Program fails, results mostly incorrect, or assumptions not understood.

Assessment Rubrics — Experiment 1: Simulation and Comparison of FDMA, TDMA and CDMA

Assessment criterion	Weight	4 — Strong	3 — Meets	2 — Partial	1 — Limited
Objective 1: Capacity estimation and comparison	10	Correct FDMA, TDMA and CDMA estimates; clear	Outputs mostly correct, with a minor error or incomplete explanation.	Some correct outputs, but errors or omissions remain.	Program fails, results mostly incorrect, or assumptions not understood.

Assessment Criterion	4 — Strong	3 — Meets	2 — Partial	1 — Limited
Parameter study and recorded results	Changes relevant parameters, reruns applicable programs, records outputs and explains effects.	Provides rerun results and comparison, but one change or explanation is incomplete.	Makes limited changes or records results without clear comparison.	No meaningful parameter study or usable results record.
Analysis and interpretation	Answers accurately, uses results as evidence and distinguishes model behavior from real-system behavior.	Most answers sound and refer to results, with some gaps in explanation.	Answers partly correct but mainly describe outputs without explaining them.	Analysis missing, unsupported or substantially incorrect.
Engineering judgment and conclusion	Gives a clear conclusion, identifies assumptions and limitations, and avoids unsupported performance rankings.	Conclusion is reasonable but omits a relevant assumption, limitation or implication.	Conclusion mostly repeats results, with little discussion of limitations.	Conclusion missing or draws unsupported conclusions.
Report organization and evidence	Includes code, tables, required figures, parameter values and clear captions.	Required evidence present, with minor organization or labeling issues.	Some evidence missing or difficult to follow.	Report substantially incomplete or unclear.
Individual explanation / viva	Clearly explains own work, formulas and results.	Explains most aspects, with minor gaps.	Gives partial explanations but struggles to connect code and results.	Cannot explain key parts of the submitted work.